Tuan Pham

AI/ML Engineer

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AI/ML engineer and physicist specializing in scalable machine learning on HPC systems. Track record of accelerating scientific data processing and simulation 10–30x on GPUs with GNNs, transformers, diffusion models and normalizing flows, and of bridging Python-based research to production C++. Hands-on with GPU benchmarking and optimization (mixed precision, JIT, quantization) across CPU/GPU HPC architectures on NERSC (Perlmutter), CC-IN2P3 and Ruche.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher **10/2025–Present**

Irène Joliot Curie Lab — CNRS/Université Paris-Saclay *Orsay, France*

- Develop AI/ML solutions for simulation and data processing in particle physics, focusing on graph neural networks, transformers, normalizing flows and diffusion models.
- Design statistical methods to improve sampling quality of generative models for physics simulation, yielding high fidelity with a 10–20x throughput increase over traditional simulation; developed and benchmarked on CC-IN2P3 and Ruche (A100/V100 GPUs).
- Enhance scalability of GNN inference on large graphs (45% less latency/GPU memory) with fine-tuning, dynamic graph pruning and optimized nearest-neighbor search; benchmarked on NERSC Perlmutter (A100 GPUs).

Doctoral researcher **01/2023–08/2025**

European Organization for Nuclear Research *Geneva, Switzerland*

- Developed Graph Neural Networks for large point-cloud processing, achieving a 30x GPU acceleration over baseline methods. Scaled and optimized models on 20 million simulated particle collisions, maintaining 99% recall and precision.
- Reduced inference latency by 50% via JIT compilation, automatic mixed precision and INT8 quantization of model parameters.
- Deployed Python-trained models into production C++ software, ensuring interoperability between research prototypes and HPC production systems.
- Built CI/CD pipelines with unit testing to ensure code quality and reproducibility across a 10-person team.

Research Fellow **01/2022–01/2023**

Lawrence Berkeley National Laboratory *Berkeley, USA*

- Developed surrogate models for detector simulation using normalizing flows and GANs.
- Boosted throughput by 70% at high fidelity by training models on 10M simulated interactions.
- Optimized the development lifecycle with online monitoring, GPU profiling and HPC job scheduling to maximize utilization and throughput.

Research Assistant **01/2020–12/2021**

European Organization for Nuclear Research *Geneva, Switzerland*

- Extracted rare physics phenomena from large-scale datasets and trained Boosted Decision Trees for event classification.
- Quantified signals and confidence intervals through statistical hypothesis testing, and developed custom analysis software in Python and C++.

EDUCATION

Ph.D. & M.S. in Physics **08/2018–08/2025**

University of Wisconsin-Madison *USA*

Doctoral Dissertation: Search for Dark Matter with the ATLAS detector and development of a track reconstruction algorithm based on Graph Neural Network for the ATLAS Inner Tracker

B.S. in Physics and Chemistry **09/2014–06/2018**

Union College *USA*

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, TensorFlow, Graph Neural Networks, Transformers, Diffusion Models, GANs, Normalizing Flows. Familiar with LLM architectures and RAG (coursework).
- **Data Science:** Scikit-learn, XGBoost, Pandas, NumPy, SciPy, statistical analysis.
- **HPC & Performance:** Python, C++, CUDA, parallel & multi-GPU training, GPU profiling, mixed precision, quantization, ONNX, Slurm, Docker/Apptainer, GitLab CI/CD, Linux, PostgreSQL.
- **Visualization:** Matplotlib, Seaborn, Plotly.
- **Languages:** English (fluent), French (fluent), Vietnamese (native).

SELECTED PUBLICATIONS

- ATLAS Collaboration. "Optimizations of the ATLAS ITk GNN reconstruction pipeline." ATL-PHYS-PUB-2025-046, CERN (2025). cds.cern.ch/record/2948192
- ATLAS Collaboration. "Computational Performance of the ATLAS ITk GNN Track Reconstruction Pipeline." ATL-PHYS-PUB-2024-018, CERN (2024). cds.cern.ch/record/2914282
- G. Aad et al. (ATLAS Collaboration). "Combination and summary of ATLAS dark matter searches interpreted in a 2HDM with a pseudo-scalar mediator using 139 fb^{-1} of $\sqrt{s} = 13$ TeV pp collision data." Science Bulletin 69 (2024) 3005. doi:10.1016/j.scib.2024.06.003
- T. Pham, X. Ju. "Simulation of Hadronic Interactions with Deep Generative Models." EPJ Web Conf. 295 (2024) 09034. doi:10.1051/epjconf/202429509034
- S. Caillou, P. Calafiura, X. Ju, D. Murnane, T. Pham, C. Rougier, J. Stark, A. Vallier. "Physics Performance of the ATLAS GNN4ITk Track Reconstruction Chain." EPJ Web Conf. 295 (2024) 03030. doi:10.1051/epjconf/202429503030
- A. Huang, Y. Melkani, P. Calafiura, A. Lazar, D. T. Murnane, T. Pham, X. Ju. "A Language Model for Particle Tracking." arXiv:2402.10239 (2024).
- H. Torres, J. Burleson, S. Caillou, P. Calafiura, J. Chan, C. Collard, X. Ju, D. Murnane, M. Neubauer, T. Pham, C. Rougier, J. Stark, A. Vallier. "Physics performance of the ATLAS GNN4ITk track reconstruction chain." Connecting the Dots Workshop (CTD 2023), Toulouse, France (2023). doi:10.5281/zenodo.15178159
- R. Liu, P. Calafiura, S. Farrell, X. Ju, D. T. Murnane, T. Pham. "Hierarchical Graph Neural Networks for Particle Track Reconstruction." arXiv:2303.01640 (2023).
- S. L. Wu, S. Sun, W. Guan, C. Zhou, J. Chan, C. L. Cheng, T. Pham, Y. Qian, A. Wang, R. Zhang, M. Livny, J. Glick, P. Barkoutsos, S. Woerner, I. Tavernelli, F. Carminati, A. Di Meglio, A. C. Y. Li, J. Lykken, P. Spentzouris, S. Y. Chen, S. Yoo, T. Wei. "Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC." Phys. Rev. Research 3 (2021) 033221. doi:10.1103/PhysRevResearch.3.033221
- N. Mann, J. Matli, T. Pham. "Semi-classical methods, angular momentum, and non-analytic problems." Eur. J. Phys. 42 (2021) 035402. doi:10.1088/1361-6404/abdb9c

TALKS & POSTERS

- "Improving neural generative models via Monte-Carlo sampling with density-ratio estimation." IN2P3/IRFU Machine Learning workshop 2026, LPSC Grenoble (Nov 2026, upcoming) (Indico).
- "A geometric deep learning algorithm for charged-particle track reconstruction in the ATLAS ITk." ICHEP 2024, Prague (Jul 2024) (Indico).
- "Graph Neural Networks for track reconstruction in the ATLAS ITk detector." AISSAI Workshop on Heterogeneous Data and Large Representation Models in Science, Toulouse (Sep–Oct 2024) (Indico).
- "Accelerating charged-particle track reconstruction with deep neural networks for point-cloud segmentation." CHEP 2026, Bangkok (May 2026) (Indico).
- "Improving simulation of extreme EM calorimeter showers with deep neural networks." CHEP 2026, Bangkok (May 2026) (Indico).
- "Improving neural generative models with density estimation and Monte-Carlo resampling." ML4Jets 2026, Vienna (Sep 2026) (Indico).
- "Heterogeneity in Graph Neural Networks for Track Reconstruction in the ATLAS Upgrade ITk Detector." Connecting the Dots 2023, Toulouse (Oct 2023) (Indico).
- "GUNTAM: A Transformer for seed reconstruction." CHEP 2026, Bangkok (May 2026) (Indico).
- "Physics Performance of the ATLAS GNN4ITk Track Reconstruction Chain." CHEP 2023, Norfolk, VA, USA (May 2023) (Indico).
- "Developments of GNN Track Reconstruction for the ATLAS ITk Detector." ACAT 2025, Hamburg, Germany (Sep 2025) (Indico).
- "First functional prototype for the reconstruction of charged particle tracks using machine learning in the ATLAS experiment at the HL-LHC." CHEP 2026, Bangkok (May 2026) (Indico).
- "Accelerating GNN Inference for the GNN4ITk Pipeline in ATLAS." CHEP 2026, Bangkok (May 2026) (Indico).

TRAINING & OUTREACH

- Mentor Master's interns in ML methodology and distributed training on HPC clusters; onboard new colleagues and review their code; ran a hands-on tutorial session at the ATLAS Inner Detector Tracking Workshop (Château de Bossey, 2023).
- Active in international collaborations (ATLAS, CERN, LBNL, IJCLab); contribute to open-source software frameworks (GNN4ITk/ACORN, ATLAS Athena).

REFERENCES

- David Rousseau — IJCLab, CNRS/Université Paris-Saclay | rousseau@ijclab.in2p3.fr
- Paolo Calafiura — Lawrence-Berkeley National Lab | PCalafiura@lbl.gov